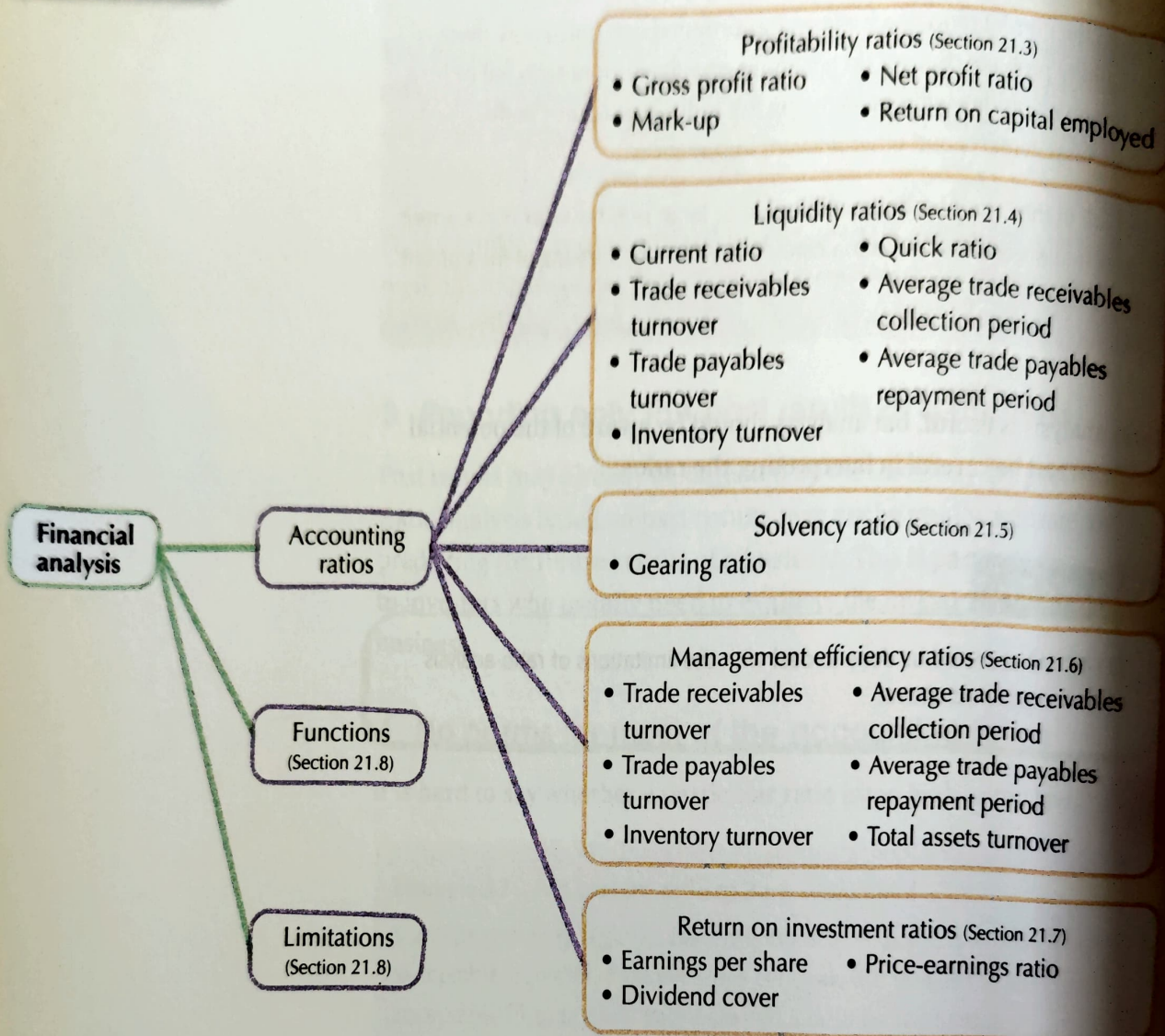


Concept Map



Glossary

- Profitability:** The ability of a business to earn profits.
- Gross profit ratio:** This measures the gross profit for every dollar in sales.
- Mark-up:** This refers to the gross profit as a percentage of the cost of goods sold.
- Net profit ratio:** This measures the net profit for every dollar in sales.
- Return on capital employed:** This measures the ability of a business to generate profits on its capital.
- Liquidity:** The ability of a business to meet its short-term financial obligations when due.
- Current ratio:** This indicates the number of times that the current liabilities of a business are covered by its current assets.
- Quick ratio:** This indicates the number of times that the current liabilities of a business are covered by its liquid assets.

- 9 **Trade receivables turnover:** This measures the size of trade receivables relative to credit sales.
- 10 **Average trade receivables collection period:** This measures the average length of time a business takes to collect trade receivables.
- 11 **Trade payables turnover:** This measures the size of trade payables relative to credit purchases.
- 12 **Average trade payables repayment period:** This measures the average length of time a business takes to settle trade payables.
- 13 **Inventory turnover:** This measures the size of inventory relative to the cost of goods sold.
- 14 **Solvency:** The ability of a business to meet its financial obligations, both short-term and long-term.
- 15 **Gearing ratio:** This indicates the ability of a business to meet long-term financial obligations.
- 16 **Management efficiency:** The ability of a business to utilise its assets efficiently.
- 17 **Total assets turnover:** This measures how efficiently assets (current and non-current) are being used to generate sales revenue.
- 18 **Earnings per share:** This measures the amount of profit earned per share issued.
- 19 **Dividend cover:** This measures the extent to which a company pays dividends out of its profits.
- 20 **Price-earnings ratio:** This measures the size of earnings per share relative to its current market price.

Summary

1 Profitability ratios include:

Ratio	Formula and interpretations	Express in
Gross profit ratio (margin)	$\frac{\text{Gross profit}}{\text{Sales}} \times 100\%$	%
	A higher gross profit ratio means that sales are more profitable.	
Mark-up	$\frac{\text{Gross profit}}{\text{Cost of goods sold}} \times 100\%$	%
	A higher mark-up means that sales are more profitable.	
Net profit ratio	$\frac{\text{Net profit before tax}}{\text{Sales}} \times 100\%$	%
	The higher the ratio, the greater the ability of the business to generate net profits from its sales.	
Return on capital employed	$\frac{\text{Profit before interest and tax}}{\text{Average capital employed}} \times 100\%$	%
	The higher the ratio, the more capable the business is in generating profits from its long-term funds.	

2 Liquidity ratios include:

Ratio	Formula and interpretations	Express in
Current ratio	$\frac{\text{Current assets}}{\text{Current liabilities}} : 1$ The higher the ratio, the more liquid the business is.	Ratio
Quick ratio	$\frac{\text{Current assets} - \text{Inventory}}{\text{Current liabilities}} : 1$ The higher the ratio, the more liquid the business is.	Ratio
Trade receivables turnover	$\frac{\text{Credit sales}}{\text{Average trade receivables}}$ When this ratio increases, it means that the business is able to collect its trade receivables more quickly.	Times
Average trade receivables collection period	$\frac{\text{Average trade receivables}}{\text{Credit sales}} \times 365 \text{ days (or 12 months)}$ The shorter the period, the faster the business is able to collect its trade receivables.	Days/ months
Trade payables turnover	$\frac{\text{Credit purchases}}{\text{Average trade payables}}$ When this ratio increases, it means that the business is able to repay its trade payables more quickly.	Times
Average trade payables repayment period	$\frac{\text{Average trade payables}}{\text{Credit purchases}} \times 365 \text{ days (or 12 months)}$ The shorter the period, the faster the business settles its trade payables.	Days/ months
Inventory turnover	$\frac{\text{Cost of goods sold}}{\text{Average inventory}}$ When this ratio increases, it means that the business is selling inventory more quickly.	Times

3 Solvency ratio includes:

Ratio	Formula and interpretations	Express in
Gearing ratio	$\frac{\text{Non-current liabilities} + \text{Preference share capital}}{\text{Non-current liabilities} + \text{Shareholders' fund}} \times 100\%$ The higher the ratio, the riskier the business is considered to be.	%

4 Management efficiency ratios include:

Ratio	Formula and interpretations	Express in
Total assets turnover	$\frac{\text{Sales}}{\text{Total assets}}$	Times
	The higher the ratio, the more efficiently the business is using its assets to generate sales revenue.	
Trade receivables turnover, average trade receivables collection period		
Trade payables turnover, average trade payables repayment period		
Inventory turnover		

5 Return on investment ratios include:

Ratio	Formula and Interpretations	Express in
Earnings per share (EPS)	$\frac{\text{Net profit after tax} - \text{Preference dividend}}{\text{Number of ordinary shares issued}}$	Amount
	The higher the ratio, the higher the potential return from holding shares in the company.	
Dividend cover (Ordinary shares)	$\frac{\text{Net profit after tax} - \text{Preference dividend}}{\text{Ordinary dividend paid}}$	Times
	The higher the ratio, the smaller the proportion of profits distributed as dividends by the company.	
Price-earnings ratio (P/E ratio)	$\frac{\text{Current market price per share}}{\text{Earnings per share}}$	Numerical value, such as 3, without any unit
	A high P/E ratio could be an indicator to sell a company's shares while a low P/E ratio could be an indicator to buy its shares.	

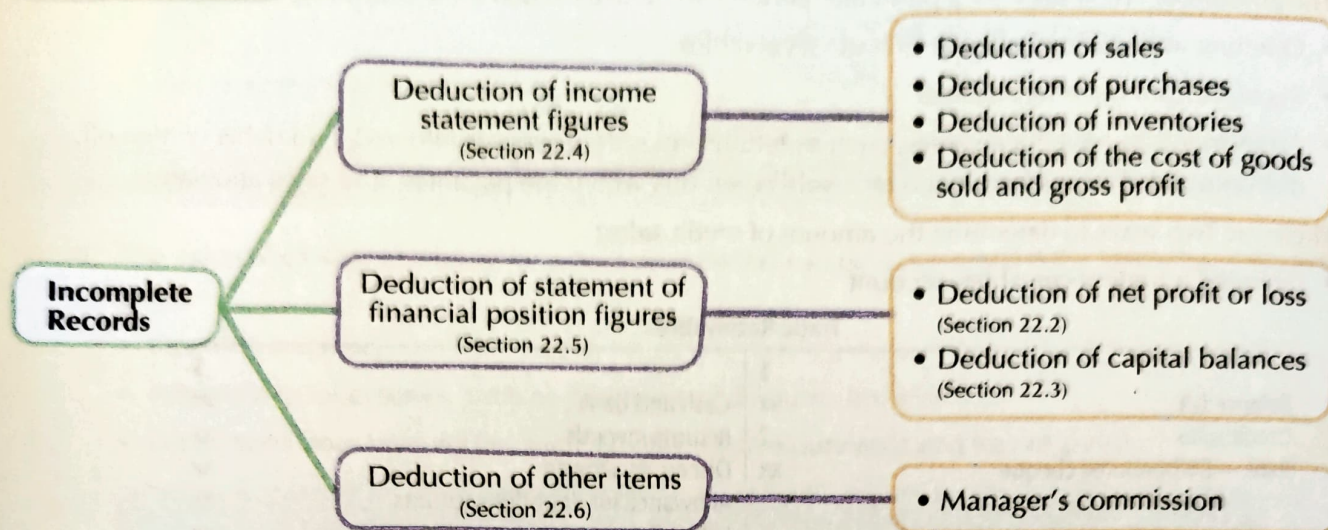
6 Functions of ratio analysis include:

- Accounting ratios express meaningful relationships between different figures in the financial statements.
- Accounting ratios calculate the relative magnitude of financial figures and therefore facilitate the comparison of financial performance.

7 Limitations of ratio analysis include:

- Ratio analysis cannot tell the whole story about a business because it fails to take into account some useful qualitative information.
- It is not fair or useful to compare the ratios of different businesses that have adopted vastly different accounting policies and methods.
- Ratio analysis based on past results may not be totally accurate for predicting the future results of a business.
- There are no norms for most of the accounting ratios. It is hard to say whether a particular ratio is too high or too low.
- Ratios may indicate the existence of a problem, but most of the time they cannot tell what the problem is or how to solve it.

Concept Map



Glossary

Statement of affairs: A statement that lists the balances of assets and liabilities of a business as at a certain date.

Summary

- The net profit or loss for a particular period can be determined as follows:

$$\text{Net profit / (loss) for the period} = \text{Closing capital} - \text{Opening capital} + \text{Drawings} - \text{Additional capital contributed}$$

- The opening and closing capital balances can be determined if information on the assets and liabilities at the end of the previous period and current period is given.
 - There are two ways to find the capital balances:
 - Using the accounting equation: $\text{Capital} = \text{Assets} - \text{Liabilities}$
 - Preparing a statement of affairs which shows the subtraction of liabilities from assets in order to calculate the capital balance.

Statement of Affairs as at [date]

Assets	\$
List of all current and non-current assets	xx
Less Liabilities:	
List of all current and non-current liabilities	xx
Capital	xx

- 3 (i) The amount of cash sales can be found from the cash book or bank statements.
- (ii) The amount of credit sales for a particular period can be determined if the following information is given:
- Opening and closing balances of trade receivables
 - Receipts from trade receivables
 - Adjustments to trade receivables, such as returns inwards, discounts allowed, bad debts written off, dishonoured cheques from trade receivables, set-offs with trade payables, and sales allowance
- (iii) There are two ways to determine the amount of credit sales:
- Opening a trade receivables account

Trade Receivables			
	\$		\$
Balance b/f	xx	Cash and bank	xx
Credit sales	?	Returns inwards	xx
Bank – Dishonoured cheque	xx	Discounts allowed	xx
		Allowance for doubtful accounts	xx
		Sales allowance	xx
		Balance c/f	xx
	xx		xx
	==		==

- Using accounting ratios:

$$\text{Credit sales} = \frac{\text{Average trade receivables} \times \text{Trade receivables turnover}}{\text{Or}} \times 365 \text{ days or 12 months}$$

net of returns inwards and sales allowance, if any

- 4 (i) The amount of cash purchases can be found from the cash book or bank statements.
- (ii) The amount of credit purchases for a particular period can be determined if the following information is given:
- Opening and closing balances of trade payables
 - Payments to trade payables
 - Adjustments to trade payables, such as returns outwards, discounts received, set-offs with trade receivables and purchase allowance
- (iii) There are two ways to determine the amount of credit purchases:
- Opening a trade payables account

Trade Payables			
	\$		\$
Cash and bank	xx	Balance b/f	xx
Returns outwards	xx	Credit purchases	?
Discounts received	xx		
Purchase allowance	xx		
Balance c/f	xx		
	xx		
	==		==

- Using accounting ratios:

$$\text{Credit purchases} = \frac{\text{Average trade payables} \times \text{Trade payables turnover}}{\text{Or}} \times \frac{\text{Average trade payables}}{\text{Average trade payables repayment period}} \times 365 \text{ days or 12 months}$$

net of returns outwards and purchase allowance, if any

- 5 (i) The value of closing inventory for a particular period can be determined if the following information is given:

- Opening inventory
- Additions to inventories, such as purchases and returns inwards
- Deductions from inventories, such as sales, returns outwards and loss of goods

$$\text{Closing inventory} = \text{Opening inventory} + \begin{array}{l} \text{Additions to} \\ \text{inventories:} \\ \text{Purchases} \\ \text{Returns inwards} \end{array} - \begin{array}{l} \text{Deductions from} \\ \text{inventories:} \\ \text{Sales} \\ \text{Returns outwards} \\ \text{Loss of goods} \end{array}$$

- (ii) As inventory is valued at cost but sales and returns inwards are stated at selling prices, these two figures have to be converted into cost in the following manner:

$$\text{Cost} = \text{Selling price} \div (1 + \text{Mark-up})$$

$$\text{Cost} = \text{Selling price} \times (1 - \text{Margin})$$

- (iii) There are two ways to determine the closing inventory figure:

- Preparing a statement to show the changes in inventory
- Using accounting ratios:

$$\text{Closing inventory} = \left(\frac{\text{Cost of goods sold}}{\text{Inventory turnover}} \times 2 \right) - \text{Opening inventory}$$

- 6 (i) The amount of cost of goods sold for a particular period can be determined if the following information is given:

- Opening and closing inventories
- Purchases and other related costs (net of returns outwards and purchase allowance, if any)

- (ii) Once the cost of goods sold is determined, the gross profit can be ascertained.

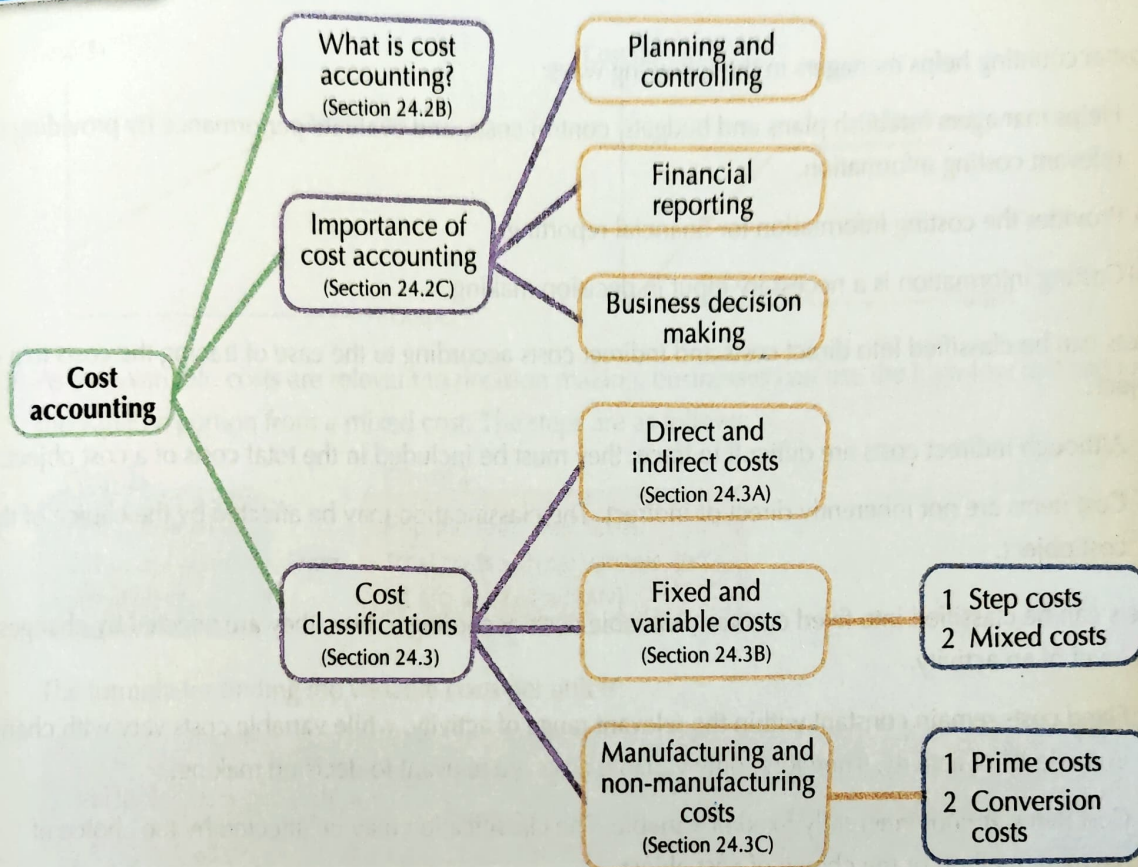
- (iii) The cost of goods sold and gross profit can also be determined using accounting ratios:

- Cost of goods sold = $\frac{\text{Gross profit}}{\text{Mark-up}}$ or Sales $\times$ (1 - Margin)
- Gross profit = Sales $\times \frac{\text{Mark-up}}{1 + \text{Mark-up}}$ or Sales $\times$ Margin

- 7 Sometimes, a firm may need to deduce the amount of the manager's commission:
- If the commission is calculated as a percentage of net profit before commission, it is simply calculated by multiplying the commission percentage with the net profit figure.
 - If the commission is calculated as a percentage of net profit after commission, it is calculated as:

$$\frac{\text{Commission percentage}}{100 + \text{Commission percentage}} \times \text{Net profit before commission}$$

Concept Map

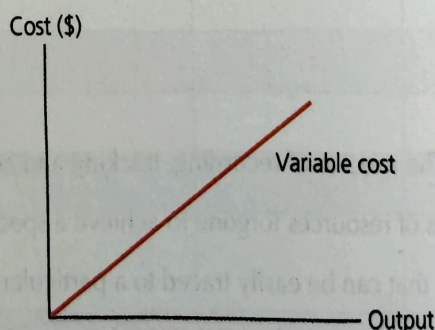
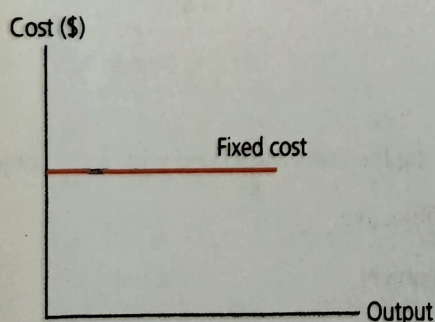


Glossary

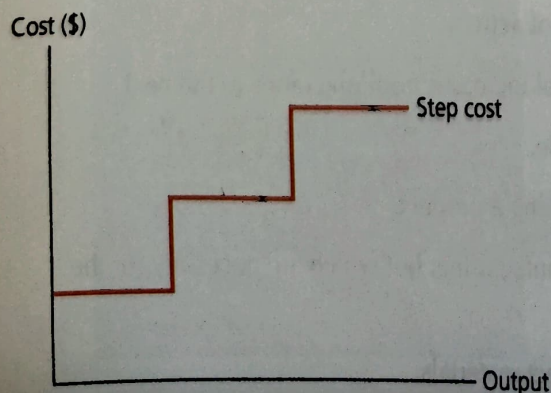
- 1 **Cost accounting:** The process of recording, tracking and analysing the costs associated with a cost object.
- 2 **Costs:** The amounts of resources forgone to achieve a specific objective.
- 3 **Direct costs:** Costs that can be easily traced to a particular cost object.
- 4 **Indirect costs:** Costs that cannot be easily traced to a particular cost object.
- 5 **Fixed costs:** Costs that remain constant within a certain range of activity level.
- 6 **Variable costs:** Costs that vary with changes in the level of activity.
- 7 **Step costs:** Costs that increase in steps as the activity level increases from one range to the next.
- 8 **Mixed costs:** Costs that are partly fixed and partly variable.
- 9 **Manufacturing costs:** Costs incurred in the course of making a product.
- 10 **Non-manufacturing costs:** Costs not associated with manufacturing, but which are necessary for the operation of a manufacturing business.
- 11 **Conversion costs:** All manufacturing costs other than direct materials.

Summary

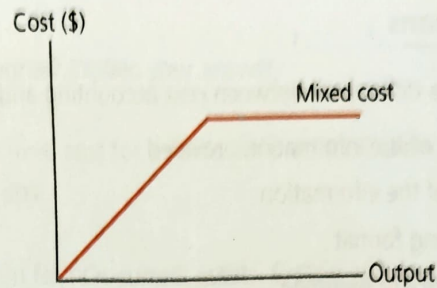
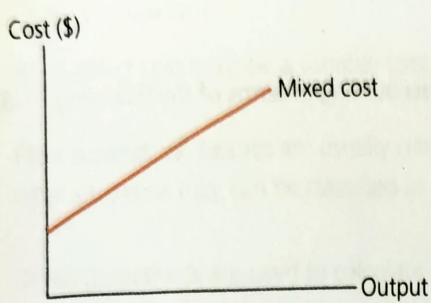
- 1 Cost accounting helps managers in the following ways:
 - (i) Helps managers establish plans and budgets, control costs, and evaluate performance by providing the relevant costing information.
 - (ii) Provides the costing information for financial reporting.
 - (iii) Costing information is a necessary input in decision making.
- 2 Costs can be classified into direct costs and indirect costs according to the ease of tracing the costs to a cost object.
 - (i) Although indirect costs are difficult to trace, they must be included in the total costs of a cost object.
 - (ii) Cost items are not inherently direct or indirect. The classification may be affected by the choice of the cost object.
- 3 Costs can be classified into fixed costs and variable costs according to how they are affected by changes in the level of an activity.
 - (i) Fixed costs remain constant within the relevant range of activity, while variable costs vary with changes in the level of activity. Therefore, only variable costs are relevant to decision making.
 - (ii) Cost items are not inherently fixed or variable. The classification may be affected by the choice of payment method or the choice of cost object.



- 4 Step costs are a type of fixed cost, but they change beyond a certain range of activity level.



- 5 (i) Mixed costs are partly fixed and partly variable. They usually change but not in direct proportion to changes in the level of activity.



- (ii) As only variable costs are relevant to decision making, businesses can use the high-low method to isolate the variable portion from a mixed cost. The steps are as follows:

Step 1

Find the variable costs per unit of activity level



Step 2

Find the total fixed costs:
Total costs – Total variable costs
(at any level of activity)

The formula for finding the variable costs per unit is:

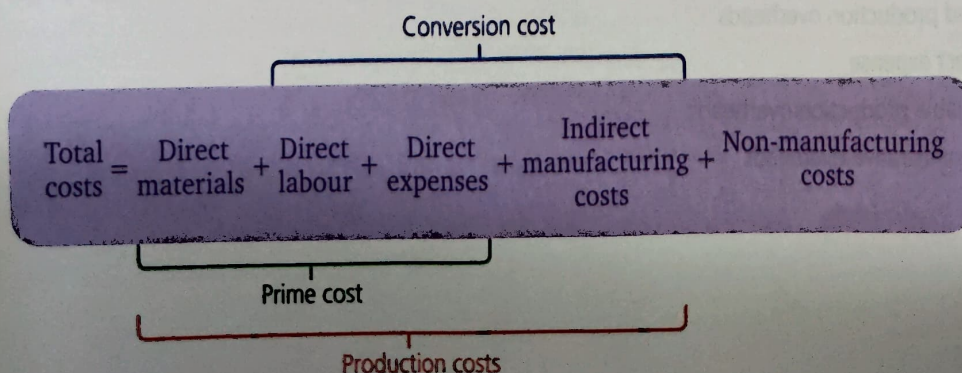
$$\text{Variable costs per unit} = \frac{\text{Costs at highest level of activity} - \text{Costs at lowest level of activity}}{\text{Highest level of activity} - \text{Lowest level of activity}}$$

- 6 Costs can also be classified by function into manufacturing costs and non-manufacturing costs.

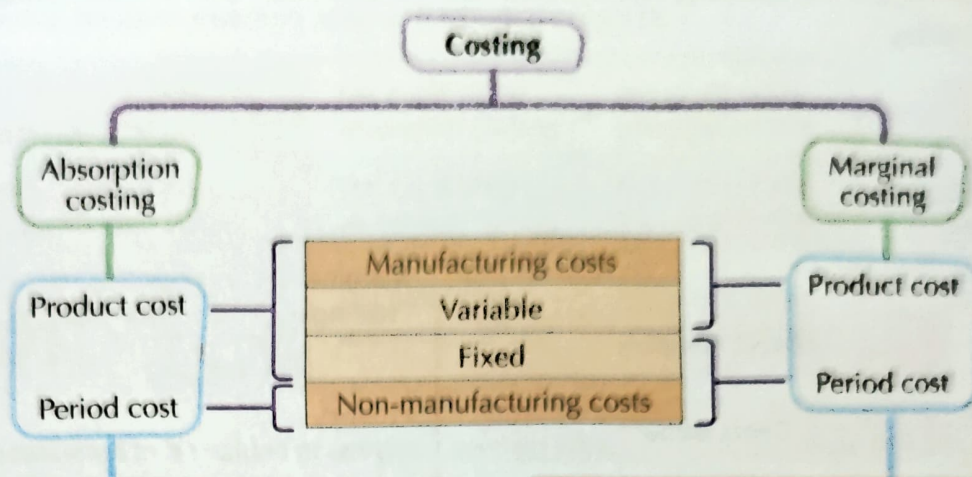
- (i) Direct manufacturing costs are manufacturing costs that can be easily traced to a particular product. The total of all direct manufacturing costs is called the prime cost.
- (ii) Indirect manufacturing costs are manufacturing costs that cannot be easily traced to a particular product.
- (iii) Non-manufacturing costs can be further classified into:

- Administrative overheads
- Selling and distribution overheads
- Research and development overheads
- Finance costs

- (iv) The calculation of the total costs of a manufacturing business is:



Concept Map



Income Statement for the year ended (date)

	\$	\$
Sales		xx
Less Cost of goods sold:		
Opening inventory	xx	
Add Cost of goods manufactured	xx	
	xx	
Less Closing inventory	xx	
	xx	
(Over-)/Under-absorbed fixed manufacturing overheads	xx	xx
Gross profit		xx
Less Non-manufacturing overheads		xx
Net profit		xx

Income Statement for the year ended (date)

	\$	\$
Sales		xx
Less Cost of goods sold:		
Opening inventory		xx
Add Cost of goods manufactured		xx
		xx
Less Closing inventory		xx
		xx
Less Variable non-manufacturing overheads		xx
Contribution margin		xx
Less Fixed manufacturing overheads	xx	
Fixed non-manufacturing overheads	xx	xx
Net profit		xx

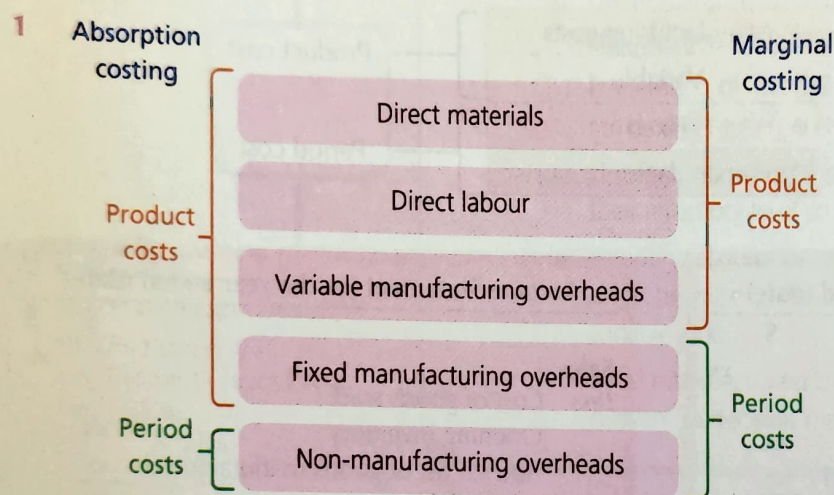
Absorption of manufacturing overheads
(Section 25.3A)

Glossary

- Product cost:** The costs of producing a product that will become the cost of goods sold when the goods are sold, or inventory when the goods remain unsold.
- Period cost:** The costs that are written off as expenses in the period when they are incurred.
- Absorption costing:** A costing system in which all manufacturing costs, either fixed or variable, are treated as product costs.
- Marginal costing:** A costing system in which only variable manufacturing costs are treated as product costs.

- 5 **Absorption base:** The basis on which overheads are absorbed into products.
- 6 **Contribution margin:** The excess of sales revenue over variable costs, both manufacturing and non-manufacturing.

Summary



- 2 (i) Under absorption costing, manufacturing overheads are usually absorbed into units of output using a predetermined overhead absorption rate.

Calculation	Budgeted overheads ÷ Budgeted quantity of absorption base
Absorption of overheads into products	Actual quantity of absorption base × Predetermined overhead absorption rate

- (ii) Any difference between overheads absorbed and actual overheads incurred in the same period should be adjusted to the cost of goods sold for the same period.

Over-absorption of overheads	To be deducted from the cost of goods sold
Under-absorption of overheads	To be added to the cost of goods sold

- 3 Comparison between the income statements prepared under absorption costing and marginal costing:

Absorption costing	Marginal costing
Costs are classified by function into manufacturing and non-manufacturing costs.	Costs are classified by cost behaviour into variable and fixed costs.
The income statement shows gross profit and net profit.	The income statement shows contribution margin and net profit.

- 4 (i) The resulting net profit figures under absorption costing and marginal costing may be different. This is due to different costs being absorbed into inventory.
- (ii) The relationship between inventory level and reported net profit:

If the inventory level increases during an accounting period	Net profit under absorption costing > Net profit under marginal costing	See Exhibits 25.2 and 25.6
If the inventory level decreases during an accounting period	Net profit under absorption costing < Net profit under marginal costing	See Exhibit 25.7 below
If the inventory level remains unchanged during an accounting period	Net profit under absorption costing = Net profit under marginal costing	

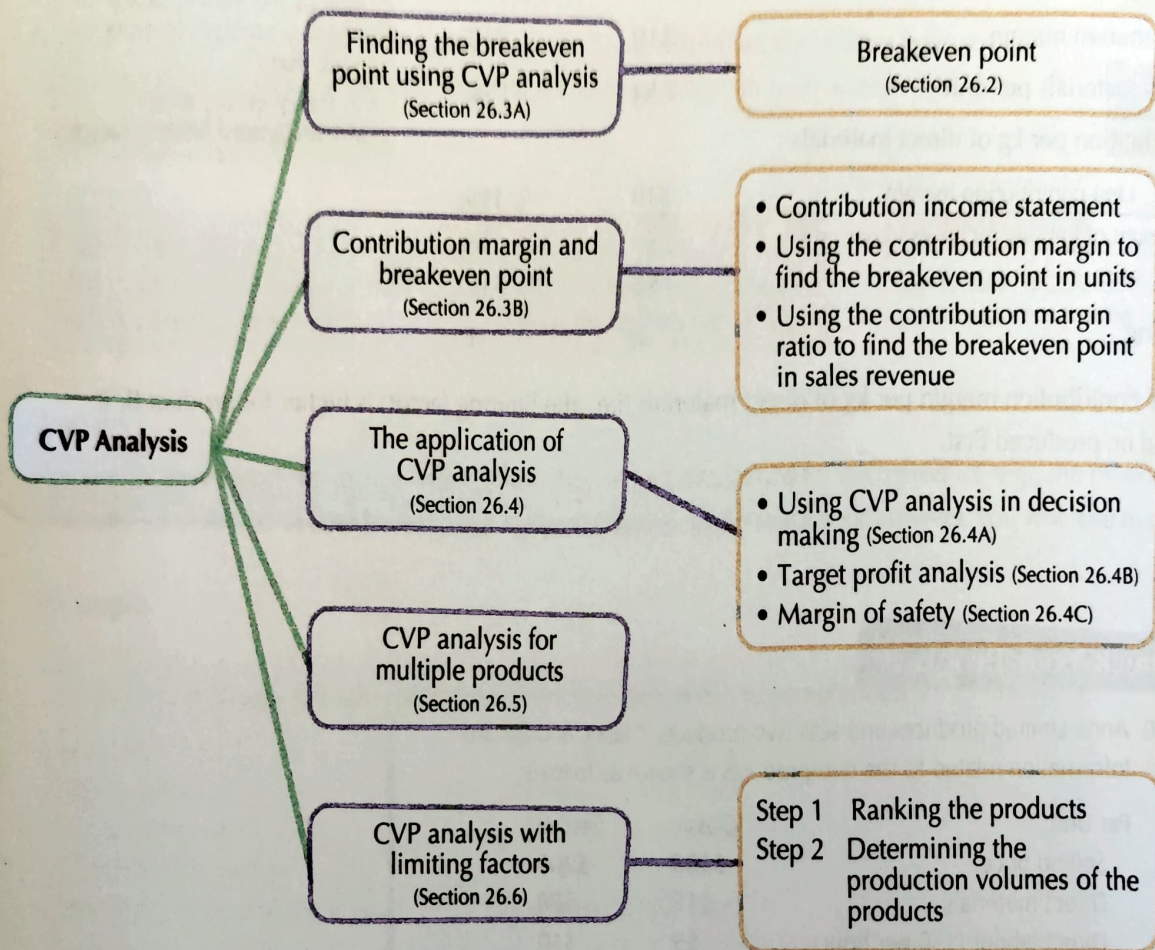
- 5 Inventories are assumed to be valued at weighted average cost.

$$\text{Weighted average cost per unit} = \frac{\text{Total cost of goods available for sale}}{\text{Total number of units available for sale}}$$

- 6 Advantages and disadvantages of adopting absorption and marginal costing:

	Absorption costing	Marginal costing
Advantages	- This approach is required in the preparation of financial statements for external reporting. - Distinguishing between manufacturing and non-manufacturing costs is easier. - Useful for pricing decisions as the price set is high enough to cover all the costs incurred in producing a product. - Including both fixed and variable manufacturing costs in inventory valuation can better reflect the costs incurred to produce goods.	- Fixed manufacturing costs are sunk costs and irrelevant to decision-making. Excluding such costs can help make better decisions. - Income statements prepared under marginal costing are more useful. They can help managers predict the changes in net profit as a result of different pricing and costing strategies.
Disadvantages	- Fixed manufacturing costs are sunk costs and irrelevant to decision-making. - The net profit figure can be manipulated by changing the inventory level.	- This approach is not allowed in the preparation of financial statements for external reporting. - Distinguishing between fixed and variable costs is difficult.

Concept Map



Glossary

- Breakeven point:** The level of activity (i.e., volume of production and sales) where total costs equal total sales revenues.
- Cost-volume-profit (CVP) analysis:** An analysis that shows the effects of different levels of production or sales on total costs and total sales revenue.
- Contribution income statement:** An income statement in which all variable costs are deducted from sales to arrive at the contribution margin.
- Target profit analysis:** This helps to calculate the sales volume or sales revenue needed to achieve a target net profit.
- Margin of safety:** The difference between actual or budgeted sales revenue and breakeven sales revenue.
- Limiting factors:** Inputs that limit the production of a product.

Summary

- 1 At the breakeven point, a business earns no profit or incurs no loss.
- 2 (i) Cost-volume-profit analysis allows us to find out how total costs and total sales revenue change with different levels of production or sales.
 (ii) It allows us to know how many units of output have to be produced and sold in order to break even. Any additional units sold would generate a profit for the entity.
 (iii) It helps managers understand the relationship between the entity's costs, volume and profit.
 (iv) Calculation of the breakeven point using the equation method:

$$\text{Sales} = \text{Fixed costs} + \text{Variable costs}$$

- 3 (i) The calculation of the contribution margin enables us to find out how a sale contributes towards covering the fixed costs of a business. It thus helps determine the breakeven point.
 (ii) The selling price will first be used to cover variable costs, with the balance (if any) being used to cover fixed costs.
 (iii) If the contribution margin is able to cover all the fixed costs, any amount remaining represents net profit.
 (iv) Calculation of the breakeven point using the contribution margin:

$$\text{Breakeven point (in units sold)} = \frac{\text{Fixed costs}}{\text{Unit contribution margin}}$$

$$\text{Contribution margin ratio} = \frac{\text{Unit contribution margin}}{\text{Unit selling price}} \times 100\%$$

$$\text{Breakeven point (in sales revenue)} = \frac{\text{Fixed costs}}{\text{Contribution margin ratio}}$$

- 4 The application of CVP analysis can be extended to calculate the sales volume or sales revenue needed to achieve a target net profit. This is called target profit analysis method.

$$\text{Units to be sold} = \frac{\text{Target net profit} + \text{Fixed costs}}{\text{Unit contribution margin}}$$

- 5 (i) Margin of safety tells by how much sales volume or sales revenue has to drop before a loss occurs.

$$\text{Margin of safety (in dollars)} = \text{Actual/Budgeted sales revenue} - \text{Breakeven sales revenue}$$

$$\text{Margin of safety (in units)} = \text{Actual/Budgeted sales units} - \text{Breakeven sales units}$$

- (ii) Margin of safety can be expressed in percentage terms.

$$\text{Margin of safety ratio} = \frac{\text{Margin of safety (in units)}}{\text{Actual/Budgeted sales (in units)}} \times 100\%$$

or

$$\frac{\text{Margin of safety (in sales revenue)}}{\text{Actual/Budgeted sales (in sales revenue)}} \times 100\%$$

- 6 CVP analysis can also be applied to multiple-product situations. The breakeven point can be found by:

- (i) the equation method; and
(ii) the weighted average method:

$$\text{Breakeven point (in units sold)} = \frac{\text{Fixed costs}}{\text{Weighted average unit contribution margin}}$$

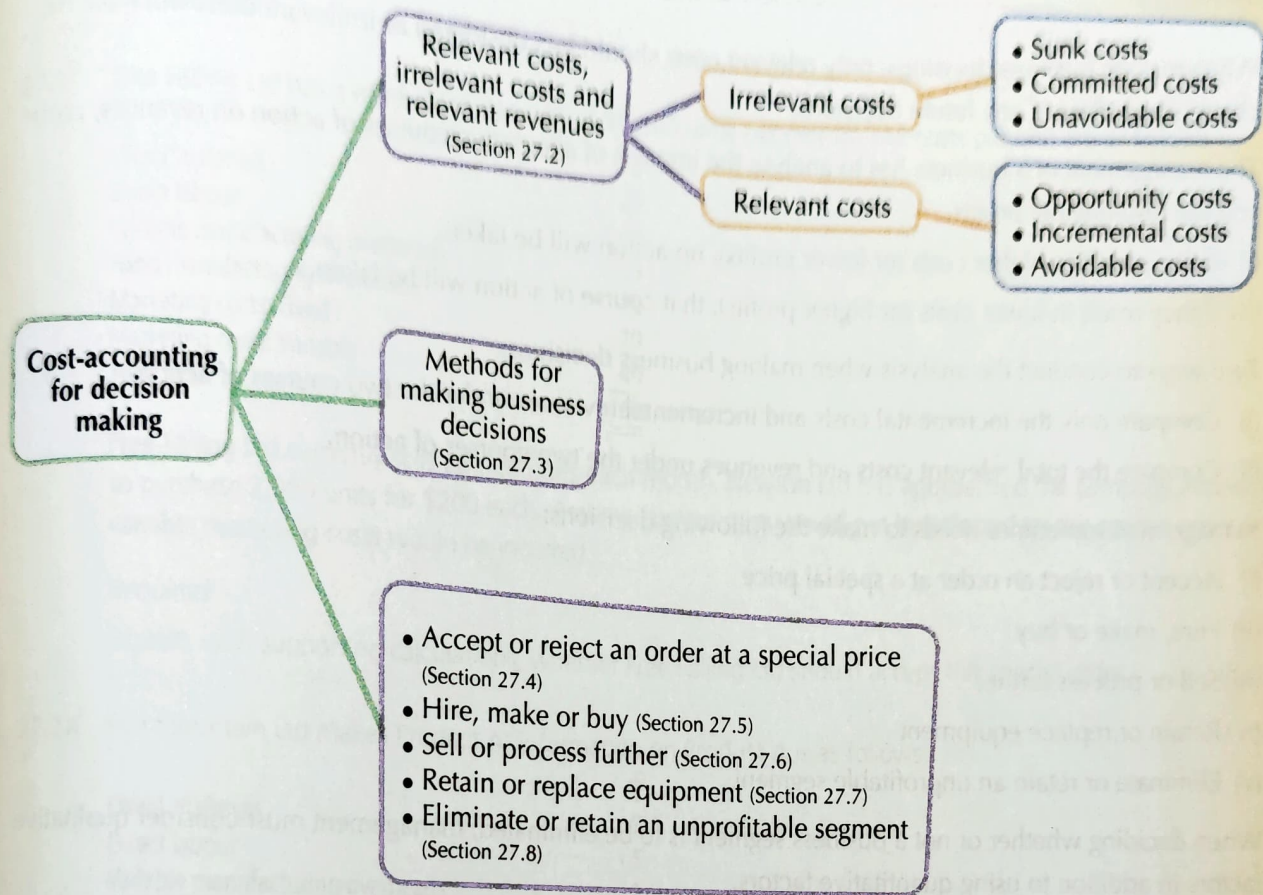
- 7 (i) When a business is faced with a limiting factor and produces more than one product, it needs to decide which product to produce first and determine the combination of production that would maximise the profit.
- (ii) When a limiting factor is present, we need to:
- Rank the products by applying the following formula:

$$\text{Contribution per unit of the limiting factor} = \frac{\text{Unit contribution margin}}{\text{Amount of limiting factor used per unit}}$$

The product with the highest contribution per unit of the limiting factor should be produced first.

- We then determine the production volume of the products.

Concept Map



Glossary

- 1 **Relevant revenues:** Expected future revenues that must be considered when making a decision.
- 2 **Relevant costs:** Expected future costs that must be considered when making a decision.
- 3 **Irrelevant costs:** Costs that do not have to be considered when making a decision.
- 4 **Sunk costs:** Costs that have already been incurred and cannot be recovered.
- 5 **Committed costs:** Costs which the firm has already undertaken.
- 6 **Unavoidable costs:** Costs that cannot be avoided or eliminated by a certain decision.
- 7 **Opportunity cost:** The highest-valued alternative forgone in making a decision.
- 8 **Incremental costs:** The additional costs incurred when a certain action is taken.
- 9 **Avoidable costs:** Costs which can be avoided or reduced when a certain action is taken.

Summary

- 1 When making business decisions, only relevant costs should be considered as irrelevant costs will not change regardless of any future decisions.
- 2 The management of a business has to analyse the impacts of alternative courses of action on revenues, costs and the resulting net profits.
 - (i) If they result in higher costs (or lower profits), no action will be taken.
 - (ii) If they result in lower costs (or higher profits), that course of action will be taken.
- 3 Two ways to conduct the analysis when making business decisions:
 - (i) Compare only the incremental costs and incremental revenues under the two courses of action.
 - (ii) Compare the total relevant costs and revenues under the two courses of action.
- 4 Management sometimes needs to make the following decisions:
 - (i) Accept or reject an order at a special price
 - (ii) Hire, make or buy
 - (iii) Sell or process further
 - (iv) Retain or replace equipment
 - (v) Eliminate or retain an unprofitable segment
- 5 When deciding whether or not a business segment is to be eliminated, management must consider qualitative factors in addition to using quantitative factors.